

INSIGHBI

Men's T-Shirt Brand Analytics

Azure SQL Database + SQL Server + Power BI Service

Azure SQL	Power BI	Interactive App
Cloud data layer	BI & DAX analysis	Published consumption

Case study objective: build a cloud-connected analytics workflow for men's T-shirt brands, clean and enrich the source data, calculate commercial metrics, compare brands, and publish an interactive Power BI App for business consumption.

Prepared as a portfolio-style data analytics case study

1. Executive Summary

This project demonstrates an end-to-end BI workflow rather than only dashboard creation. The source men's T-shirt data was loaded into Azure SQL Database, cleaned with SQL, connected to Power BI, transformed with Power Query, enriched with DAX measures/columns, and published through Power BI Service as an app.

The analysis focuses on four commercial questions: which brands offer the highest discounts, which brands carry the widest variety, which brands command higher selling prices, and which brands show stronger or weaker profit percentages.

Key finding	Observed result	Business meaning
Average discount across the analysed data	31.81%	Discounting is material and should be monitored against margin.
Highest average discount among displayed top 10 brands	52%	One leading brand is positioned with substantially deeper discounting.
Highest average sales price shown	≈ 6.1K	Premium-priced brands can be segmented separately from value brands.
Largest brand variety	51 titles / varieties	The Indian Garage has the broadest assortment among the displayed brands.
Highest average profit % shown	17%	The leading displayed brand has a 17% average profit percentage.
Lowest average profit % shown	2%	Bottom-performing brands indicate a potential margin-pressure segment.

Important data-quality note: the project uses a randomly generated profit percentage (2–17%) for dashboard analysis. Therefore, the profit findings are suitable as a BI/portfolio demonstration, but should not be presented as actual company profitability without replacing the random value with a real cost/margin field.

2. Business Problem

The raw men's T-shirt catalogue contains pricing and brand information, but raw records may include missing values and malformed price fields. A business user needs a single interactive view to compare brands and identify pricing, assortment, discount and profitability patterns.

Problem statement: How can the T-shirt catalogue be transformed into a reliable, cloud-connected BI solution that helps decision-makers compare brand pricing, discounting, assortment breadth and profitability, while handling incomplete original-price data?

3. Analytical Approach

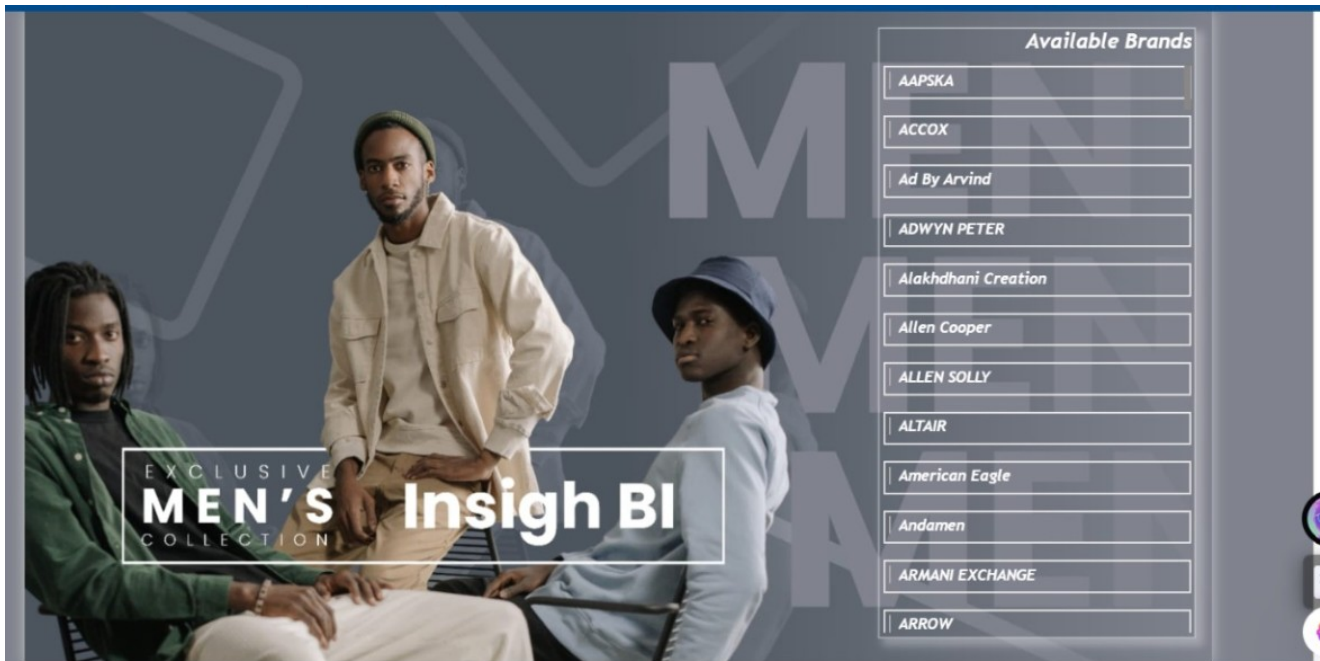
Stage	What was done	Purpose
1. Cloud setup	Created Azure Free Account, Azure SQL Database and SQL Server; configured firewall/network access.	Create a cloud-hosted analytical data source.
2. Data ingestion	Imported men's T-shirt data into SQL Server / Azure SQL.	Centralize the source data.
3. SQL cleaning	Removed '?' characters from original_price and sale_price using TRIM/REPLACE updates.	Standardize price fields before BI analysis.
4. Power BI connection	Connected Azure SQL Database to Power BI using organizational authentication.	Enable cloud-connected reporting.
5. Data transformation	Handled missing sales prices and created a factor-based fallback for missing original prices.	Create a usable Marked Price field.
6. DAX enrichment	Created Discount %, Profit %, and Cost Price calculations.	Generate business KPIs for comparison.
7. Dashboard design	Built brand selector and visual pages for discount, variety, sales price and profit.	Make insights interactive and decision-ready.
8. Deployment	Published the report to Power BI Service and created a Power BI App.	Share the analysis with end users.

Core calculations used

- **Discount %** = $(\text{Marked Price} - \text{Sales Price}) \div \text{Marked Price} \times 100$
- **Profit %** = RANDBETWEEN(2,17) in this portfolio dataset
- **Cost Price** = $100 \times \text{Sales Price} \div (100 + \text{Profit \%})$
- **Marked Price fallback**: when Original Price is missing, Sales Price $\times 1.5$ was used, representing a 50% uplift.

4. Power BI App — Brand Overview

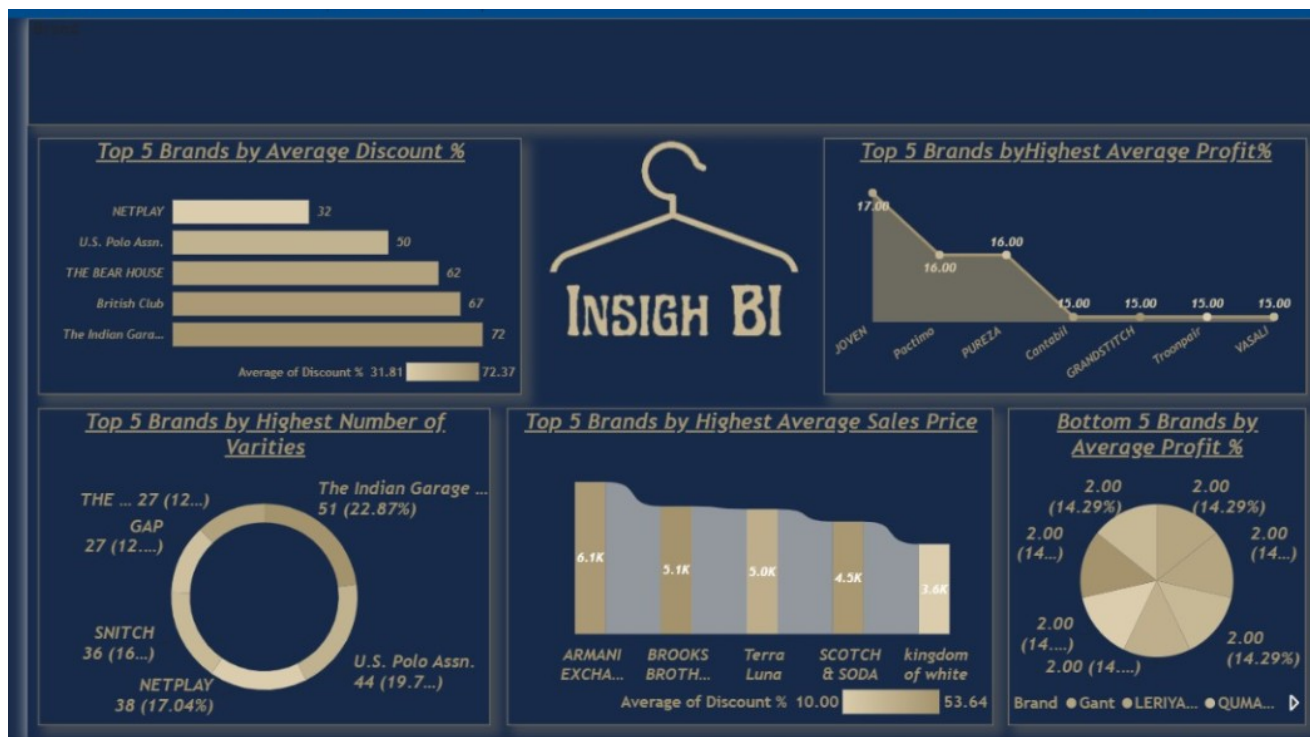
The first app page is designed as a brand navigation/selection layer. It exposes the available brands and provides a visual entry point into the detailed analysis.



Design takeaway: the app structure separates brand discovery from analytical detail, making it easier for a user to select a brand and then inspect commercial metrics.

5. Power BI App — Brand Performance Analysis

The detail page brings together five comparison views: top brands by average discount, top brands by average profit percentage, top brands by number of varieties, top brands by average sales price, and bottom brands by average profit percentage.



The visual layer converts row-level product data into brand-level comparisons that can support pricing and assortment decisions.

6. Quantified Findings

The following results are read directly from the supplied Power BI screenshots. Values shown as approximately 6.1K, 5.1K, etc. are reported at the dashboard's displayed precision.

Discounting

Result: The dashboard shows an overall average discount of 31.81%. The five displayed leading brands have average discounts of approximately 72%, 67%, 62%, 50% and 32%.

Business implication: High-discount brands should be evaluated for whether discounting is driving volume or eroding margin.

Assortment breadth

Result: The Indian Garage leads the displayed variety chart with 51 titles/varieties (22.87%), followed by U.S. Polo Assn. with 44 and Netplay with 38.

Business implication: The Indian Garage has the broadest displayed assortment, giving it more product variety to serve different customer preferences.

Sales-price positioning

Result: The five displayed highest average sales prices are approximately 6.1K, 5.1K, 5.0K, 4.5K and 3.6K.

Business implication: These brands can be treated as a premium-price segment and compared separately from lower-priced brands.

Profitability

Result: The highest displayed average profit percentage is 17%, followed by brands at 16% and 15%. The bottom-5 visual shows 2% average profit for each displayed bottom brand.

Business implication: There is a visible 15-percentage-point gap between the displayed highest and lowest profit levels.

7. What Was Analysed

Analysis area	KPI / dimension	Question answered
Brand discounting	Average Discount %	Which brands rely most heavily on discounting?
Product assortment	Count of Title / varieties	Which brands offer the widest selection?
Price positioning	Average Sales Price	Which brands command higher selling prices?
Profitability	Average Profit %	Which brands show stronger / weaker margins?
Brand exploration	Brand slicer / selector	Can users drill into a selected brand?

8. Business Recommendations

- **Review deep-discount brands:** compare the 72%, 67% and 62% discount segments with actual unit sales and contribution margin before continuing aggressive promotions.
- **Segment premium brands:** the brands with ≈6.1K–4.5K average sales prices can be analysed as a premium cohort to understand whether higher prices are supported by assortment and demand.
- **Use assortment breadth strategically:** The Indian Garage's 51 displayed varieties suggest an opportunity to analyse which product types contribute most to sales rather than simply increasing SKU count.
- **Protect low-margin brands:** brands at the displayed 2% average profit level should be reviewed for pricing, procurement cost, discounting and product mix.
- **Add time-based analysis:** monthly sales, units sold, revenue, return rate and promotion periods would allow management to distinguish sustainable performance from one-time pricing effects.

9. Data Quality & Modelling Considerations

- The original and sale price fields contained '?' characters, which were removed in Azure SQL before downstream reporting.
- Missing Original Price values were handled with a business-rule fallback: Marked Price = Sales Price × 1.5.
- Profit % was generated randomly between 2 and 17. This is a modelling/demo assumption, not an actual financial measure.
- For a production dashboard, profit should be calculated from verified revenue, cost of goods sold, discounts, returns and other relevant cost components.
- Top-N visual results should be interpreted as the brands displayed in the supplied report; the underlying full dataset was not provided with this case-study request.

10. Tools & Skills Demonstrated

Azure	SQL	Power BI
Azure Free Account Azure SQL Database Server & firewall configuration	SQL Server Management Studio SQL authentication TRIM / REPLACE / UPDATE Data cleaning	Power Query DAX Calculated columns Top-N filters Interactive visuals Power BI Service Power BI App

11. Portfolio Outcome

This project demonstrates the ability to move from cloud data storage and SQL cleaning to Power BI transformation, KPI development, visual analysis and business-facing deployment. The final Power BI report was published to Power BI Service and packaged as a Power BI App, turning the analysis into a consumable reporting solution.

Resume-ready project theme: End-to-end Azure SQL & Power BI analytics solution for men's T-shirt brand pricing, discount, assortment and profitability analysis.

Source: supplied project workflow and Power BI screenshots. Quantified dashboard values are reported at the precision visible in those screenshots.